# The Conflict Framing Elasticity:

# A Three-Layer Diagnostic of Media Coverage

# and Democratic Erosion in 40 Countries, 2017–2026

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**Abstract**

Hannah Arendt (1951) warned that the precondition for authoritarian rule is the destruction of a shared factual world. Decades later, the global news ecosystem has been algorithmically restructured in ways that warrant systematic measurement. We propose a three-layer diagnostic framework for the health of media coverage across democracies, applied to 872 million news events in the Global Database of Events, Language, and Tone (GDELT) from 1979 to 2026. We first document a severe but previously uncharacterized measurement discontinuity at the GDELT v1-to-v2 transition (2013–2015): mean tone collapses by 6.47 points while the Goldstein conflict–cooperation scale remains stable. A synchronicity diagnostic confirms this as a methodological artifact: all nine countries examined and all seventeen CAMEO event types decline identically in 2015. Within the consistent post-2015 regime, we propose three complementary measures. Layer 1 is the conflict framing elasticity ($\beta$), the regression slope of tone on the Goldstein scale, which captures how reactively media frame conflict events. Layer 2 is the position in tone×Goldstein space, characterized through a 1-skeleton graph that reveals how regime type sorts vertically along the tone axis. Layer 3 is the rate of change ($\Delta\beta$ per year), which identifies countries deteriorating fastest. Across 40 countries, $\beta$ increases significantly within the consistent regime (slope = 0.004/yr, $p = .022$) and varies meaningfully across countries (ANOVA $F = 10.64$, $p < .0001$). Seven countries show statistically significant rising $\Delta\beta$ ($p < .05$) and one is marginally significant: Honduras (+0.020/yr, $p = .016$), Uganda (+0.015/yr, $p = .001$), Malaysia (+0.013/yr, $p = .008$), Tanzania (+0.013/yr, $p = .047$), Bolivia (+0.012/yr, $p = .011$), Turkey (+0.010/yr, $p = .006$), Mexico (+0.005/yr, $p = .026$), and Ecuador (+0.015/yr, $p = .067$). These are concentrated in hybrid regimes and flawed democracies—the contested middle of Huntington's third wave—not in authoritarian regimes (which are stable or declining at the floor) or full democracies. The conflict framing elasticity provides a media-side leading indicator of democratic erosion that operates on a faster time scale than formal regime classifications, and complements existing democracy indices.

*Keywords:* GDELT, conflict framing elasticity, topological data analysis, democratic recession, Arendt, Huntington, algorithmic amplification

# The Conflict Framing Elasticity: A Three-Layer Diagnostic of Media Coverage and Democratic Erosion in 40 Countries, 2017–2026

Hannah Arendt (1951) argued that the precondition for totalitarian rule is not ideology but the destruction of a shared factual world—the creation of conditions in which citizens inhabit separate, incompatible realities. She was writing about radio and print propaganda. Her insight carries forward with precision into the algorithmic media environment of the twenty-first century: if the infrastructure through which citizens encounter political information is restructured to amplify conflict, reward outrage, and fragment audiences into factional universes, the result is not merely a change in media tone but an erosion of the communicative foundations that democratic self-governance requires.

This paper investigates that erosion empirically through a three-layer framework applied to the Global Database of Events, Language, and Tone (GDELT). Layer 1 measures conflict framing elasticity (β), the slope at which media tone responds to the conflict–cooperation intensity of reported events. Layer 2 places countries in the topological landscape of tone and Goldstein scores, revealing how regime type sorts vertically along the tone axis. Layer 3 computes the rate of change in β across 2017–2026, identifying which media environments are deteriorating, stabilizing, or recovering. Together, these three layers offer a media-side diagnostic of democratic discourse that is independent of, and complementary to, existing democracy indices.

Before any of this is credible, we must address a foundational challenge: GDELT contains a measurement discontinuity of historic magnitude that has gone formally undocumented despite the database's widespread use in academic research. Characterizing this artifact is not housekeeping; it is the necessary foundation for any longitudinal analysis of media tone and constitutes the first of this paper's contributions. The synchronicity diagnostic we propose—showing that all countries and all event types decline identically in 2015—provides a reusable test for similar artifacts in other large-scale datasets.

The paper's most politically consequential finding emerges from Layer 3. The countries whose media environments are deteriorating fastest are not the most authoritarian; they are hybrid regimes and flawed democracies in transition. Honduras, Uganda, Ecuador, Malaysia, Tanzania, Bolivia, Turkey, and Mexico stand out as the canary cases of the third wave's reversal. Authoritarian regimes are either stable or declining at the floor of the measure, consistent with state-controlled media producing uniform coverage that does not respond to changes in event intensity. The conflict framing elasticity therefore appears to be a leading indicator of democratic backsliding, operating on a faster time scale than formal regime classifications.

## Three Waves, Three Media Revolutions, Three Reversals

### *The Pattern*

Samuel Huntington's (1991) account of three waves of democratization—the long first wave (1828–1926), the post-war second wave (1943–1962), and the third wave from 1974—has become the canonical framework for understanding democratic expansion and contraction. Less commonly noted is that each wave coincided with a transformation in communicative infrastructure, and each reversal was preceded by the capture or collapse of that infrastructure.

The first wave's expansion was underwritten by the mass circulation press, which created the national public sphere that Anderson (1983) described as the "imagined community." Its partial reversal was driven by the weaponization of radio. Adena et al. (2015) provide causal evidence that Nazi radio programming significantly increased anti-Semitic sentiment and electoral support. Roosevelt's Fireside Chats demonstrated the same medium's democratic potential (Brown, 2004). The lesson was that radio's political consequences depended on the institutional framework surrounding it—and societies responded with broadcast regulation, licensing, and public media mandates.

The second wave accompanied television's spread, which created shared national experiences while initiating the erosion of local news ecosystems. Gentzkow (2006) showed that television's introduction reduced voter turnout by displacing local newspapers. Norris (2000) articulated a "virtuous circle" thesis dependent on the continued existence of quality journalistic institutions. Throughout the broadcast era, the GDELT data we analyze reflect a constrained equilibrium: stable mean tone near +5.0 from 1979 through 2011, with moderate variance.

The third wave coincided with internet optimism. Van Alstyne and Brynjolfsson (1996) warned of "balkanization." Sunstein (2001) elaborated echo chamber theory. The mechanism through which fragmentation would be realized—algorithmic curation optimized for engagement rather than informational value—was not fully anticipated.

### ***The Democratic Recession and Its Communicative Substrate***

Larry Diamond (2015) coined "democratic recession" to describe the post-2006 decline in global freedom scores. Levitsky and Ziblatt (2018) argued that contemporary erosion operates through incremental degradation of norms by elected leaders. Mounk (2018) identified a disconnect between liberal and democratic values. Lührmann and Lindberg (2019) documented a "third wave of autocratization." These accounts focus on political actors while undertheorizing the communicative infrastructure through which they operate.

Brady et al. (2017) demonstrated that morally outraged content diffuses more rapidly through social networks. Crockett (2017) theorized a feedback loop amplifying moral outrage by reducing its costs while increasing its rewards. The structural consequence is a bias toward negative, conflict-oriented content in algorithmically mediated environments—a bias that should manifest precisely as an increase in conflict framing elasticity.

Arendt's analysis becomes relevant in a specific way: she argued totalitarianism required not ideological uniformity but the dissolution of the distinction between fact and fiction. The algorithmic fragmentation of discourse produces precisely this condition—not a single dominant narrative but a kaleidoscope of incompatible narratives, each reinforced within its own factional cluster. Where the broadcast era produced a constrained equilibrium of shared reference points, the algorithmic era produces a measurable deterioration of those reference points—which the framework we propose attempts to quantify.

## Data and Methods

### ***GDELT Event Database***

We analyze the complete GDELT Event Database (Leetaru & Schrodt, 2013) via Google BigQuery (gdelt-bq.full.events), encompassing 872,575,515 news events from 1979 to early 2026. Two measures are extracted: Average Tone (sentiment score from automated content analysis; pre-2012 mean = +5.37, SD = 2.43; post-2015 mean = −1.11, SD = 4.16) and the Goldstein Scale (conflict–cooperation, −10 to +10; stable across periods at 0.58 vs. 0.53). Events are disaggregated by ActionGeo_CountryCode for 40 countries (representing all democratic regime types from full democracy to authoritarian) and by CAMEO EventRootCode for 17 event types.

### ***Synchronicity Diagnostic***

To distinguish real tonal shifts from measurement artifacts, we identify the year of maximum annual tone decline for each country and event type. Universal synchronicity—all entities shifting at the same year—confirms a methodological artifact rather than substantive media change.

### *Z-Score Correction*

To remove the level shift while preserving cross-country variation, we compute within-year z-scores: z(country, year) = [tone(country, year) − tone(global, year)] / SD(global, year). This measures how many standard deviations above or below the global annual mean a country's coverage falls, irrespective of absolute tone level.

### *Layer 1: Conflict Framing Elasticity (β)*

β is defined as the weighted ordinary least squares slope of tone on the Goldstein scale, computed per country–year cell from binned scatter data weighted by event count. β captures how many points of tone change per unit of conflict—the reactivity of media framing to event intensity. Higher β indicates a media environment that more sharply differentiates negative from positive coverage along the conflict dimension. Pre-2012 and post-2016 are treated as separate measurement regimes; analyses are restricted to post-2016.

### *Layer 2: Regime Position in Tone×Goldstein Space*

For each country we compute the weighted mean tone and mean Goldstein score during a reference year (2017–2024) and construct a 1-skeleton graph over standardized (tone, Goldstein) coordinates using a Rips filtration at threshold $\varepsilon = 1.0$. This captures which countries occupy adjacent positions in the coverage landscape—which countries, in effect, sound alike in news coverage regardless of their actual politics. We characterize the resulting topology using persistent homology (Edelsbrunner et al., 2002; Carlsson, 2009), computing $H_0$ (connected components) and $H_1$ (loops) for the pre-2012 and post-2016 regimes separately.

### *Layer 3: Rate of Change (Δβ)*

For each country, we compute the linear regression slope of annual β from 2017 through 2026 (10 observations). The slope Δβ/yr captures whether media conflict-reactivity is increasing, stable, or decreasing. We test significance via linear regression p-value and report alongside the EIU Democracy Index 2024 classification (full democracy, flawed democracy, hybrid regime, authoritarian).

### *Statistical Tests*

Post-2016 β trend: linear regression testing slope $\neq 0$. Cross-country variation: one-way ANOVA. Layer correlations with Democracy Index: Spearman rank. Pre/post regime comparison: Welch's t-test (acknowledged as confounded by measurement regime change). Supplementary figures provide additional detail: country-level β trajectories (Figure S1), topic-level coupling (Figure S2), and persistent homology diagrams (Figure S4).

## Results

### *The GDELT Tone Discontinuity*

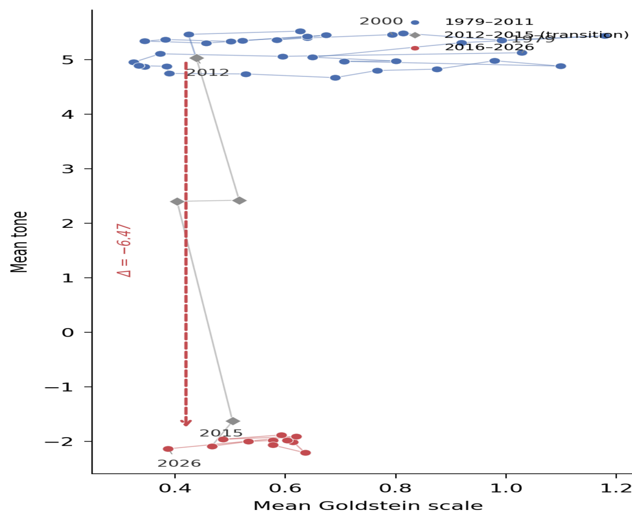


***Figure 1.*** *Annual centroid trajectory in tone×Goldstein space (1979–2026). Two distinct regimes are connected by a 2013–2015 transitional bridge. Tone shifts 6.47 points; Goldstein shifts 0.05.*

Figure 1 reveals two distinct clusters in the annual centroid trajectory of tone and Goldstein, connected by a narrow 2013–2015 bridge. The synchronicity diagnostic (Table 1) confirms this is a measurement artifact rather than substantive change: the maximum annual tone decline occurs in 2015 for all nine countries examined, ranging from −3.60 (Brazil) to −5.32 (Turkey). The timing aligns with GDELT 2.0’s launch on February 19, 2015, which introduced machine translation of 65 languages, expanded non-Western source monitoring, and deployed 24 new emotional measurement packages (GDELT Project, 2015). The topic-level diagnostic confirms the pattern: all seventeen CAMEO event types show identical cliff-edge declines in 2015. β itself exhibits a level shift at the boundary (pre-2012 mean = 0.099; post-2016 mean = 0.285; Welch’s $t = -27.88$, $p < 10^{-15}$). We therefore treat the two regimes as incommensurable and restrict all subsequent analyses to the post-2016 window.

***Table 1***

*Year of Maximum Annual Tone Decline by Country (synchronicity diagnostic)*

| **Country** | **Year of Max Decline** | **Δ Tone** |
|---|---|---|
| Turkey | 2015 | −5.32 |
| Venezuela | 2015 | −4.81 |
| Colombia | 2015 | −4.65 |
| Russia | 2015 | −4.32 |
| Mexico | 2015 | −4.17 |
| India | 2015 | −3.87 |
| United States | 2015 | −3.83 |
| United Kingdom | 2015 | −3.74 |
| Brazil | 2015 | −3.60 |

The likely mechanism is source composition. GDELT 2.0’s expansion to 65 machine-translated languages and its massive increase in non-Western source monitoring introduced sources from conflict zones and authoritarian contexts that likely carry lower average tone. The expansion also changed the denominator: events that were previously underrepresented (because they appeared only in non-English local media) gained coverage, shifting the aggregate tone downward. This source-composition hypothesis warrants direct testing through source-level filtering, which we flag as a priority for future work.

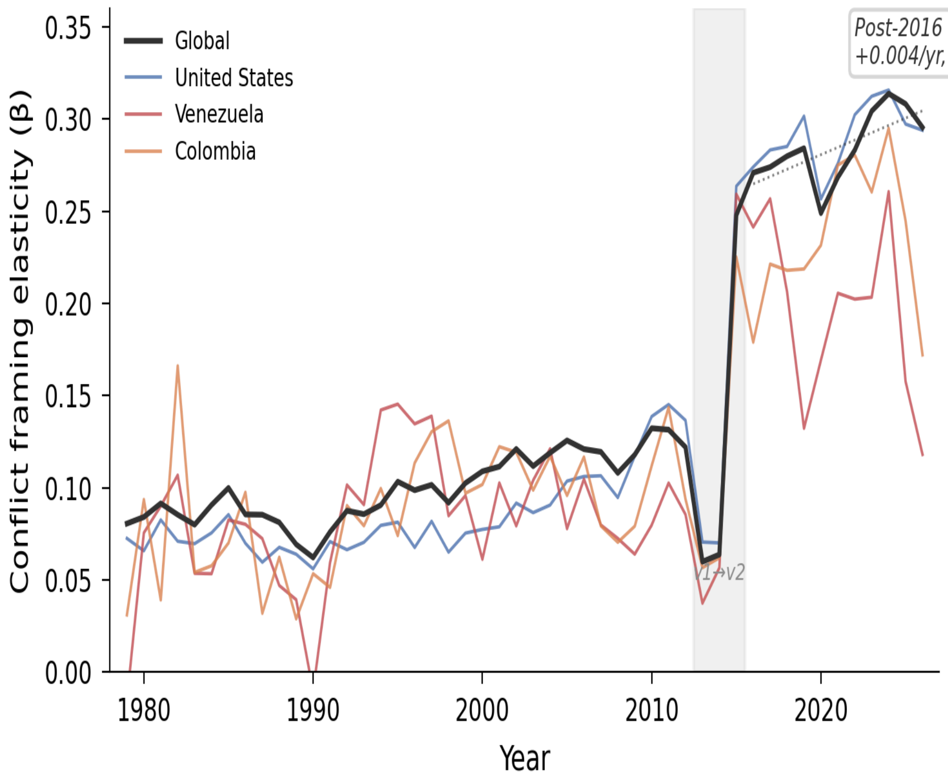


***Figure 2.*** *Global conflict framing elasticity (β) over time, with Venezuela, United States, and Colombia highlighted. Gray band: GDELT v1→v2 transition. Dotted line: post-2016 regression (slope = 0.004/yr, p = .022). Even within the consistent measurement regime, β rises significantly.*

Within the post-2016 regime, global β rises from 0.25 to 0.31—a statistically significant trend (slope = 0.004/year, $R^2$ = 0.46, p = .022). This represents a 25% increase in conflict-negativity coupling within a period of stable measurement: media environments globally are becoming more reactively responsive to conflict in their tonal framing, consistent with the algorithmic amplification thesis (Brady et al., 2017; Crockett, 2017). This is the first global-scale, longitudinal quantification of the trend.

***Layer 1: Conflict Framing Elasticity Across 40 Countries***

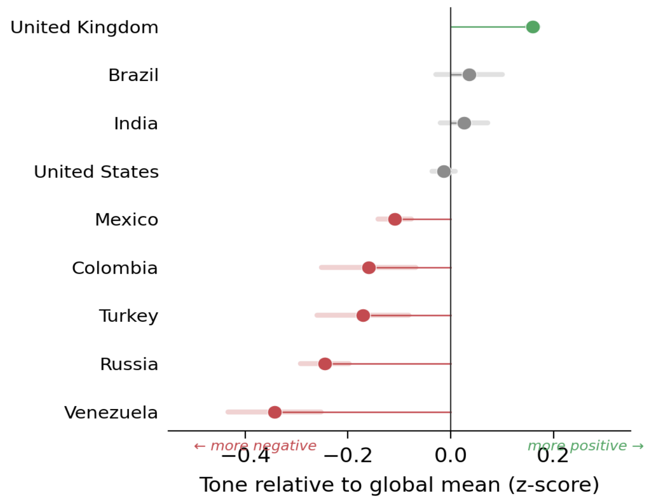


***Figure 3.*** *Z-score corrected tone by country (post-2016), ranked. Venezuela (z = −0.34) is most negative globally; United Kingdom (z = +0.16) is most positive. Error bars show 95% confidence intervals.*

Once the global level shift is removed via z-score correction, substantively plausible country-level variation emerges (Figure 3). Venezuela receives the most negatively toned coverage globally ($z = -0.34$), followed by Russia (−0.25) and Turkey (−0.17). The United Kingdom (+0.16) is the most positively positioned. These rankings are temporally stable and align with prior knowledge of these countries' political contexts.

Among the six countries with full longitudinal scatter data (1979–2026), β varies significantly (ANOVA $F = 10.64$, $p < .0001$); the variation is also evident across the broader 40-country sample. Venezuela has the lowest post-2016 mean β (0.191) while Peru has the highest (0.460). A notable inversion emerges: Venezuela has the most negative coverage overall but the lowest conflict framing elasticity, while the United States has near-global-mean coverage and high β. This distinction—between ambient negativity and conflict reactivity—is invisible in raw tone analysis.

A floor-effect analysis sharpens the interpretation. The tone gap between conflictual events (Goldstein < −2) and cooperative events (Goldstein > 2) is 2.20 points for Venezuela vs.

3.54 for the United States. Venezuela's overall tone SD (3.40) is lower than the US (3.92). Venezuela's low β reflects both a genuine framing pattern (uniformly negative coverage regardless of event type) and statistical compression from a narrower tonal range. In Arendt's terms: in environments of pervasive crisis, the distinction between ordinary and extraordinary dissolves—even cooperative events are framed through a lens of ambient negativity. The two interpretations are not mutually exclusive; the compression itself reflects the substantive condition.

### *Layer 2: Regime Position in the Tone×Goldstein Landscape*

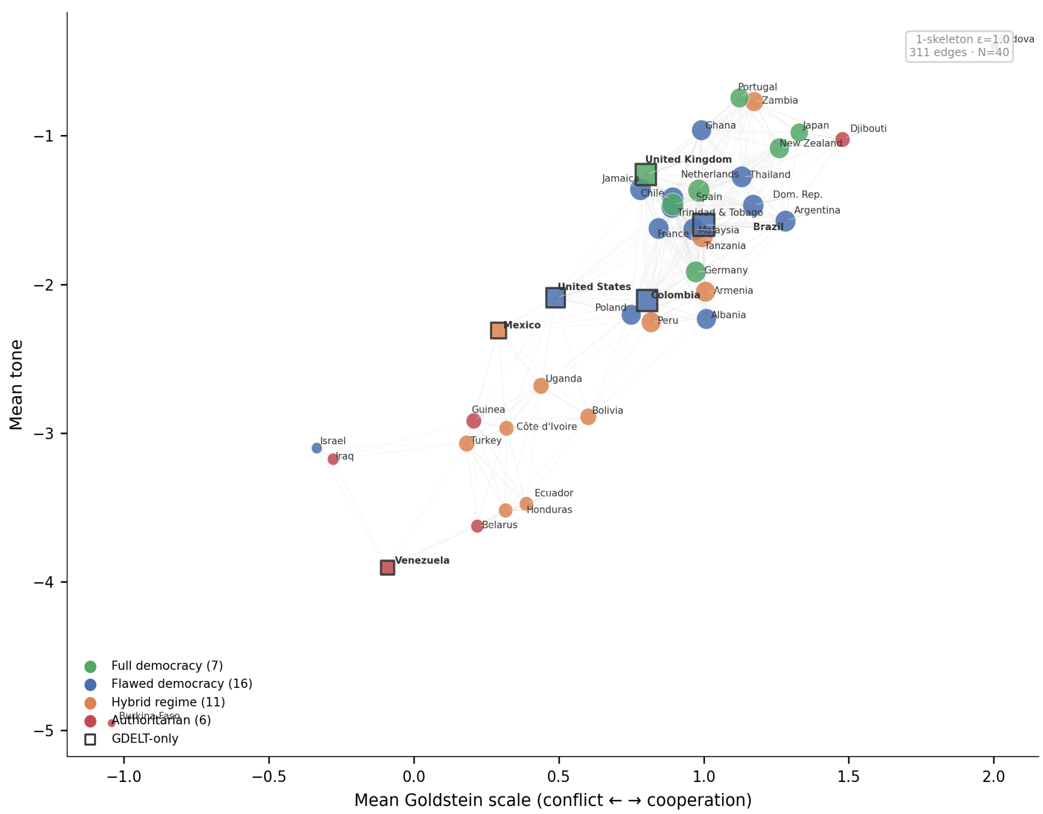


***Figure 4.*** *1-skeleton graph of 40 countries in standardized tone×Goldstein space at ε = 1.0. Node color indicates EIU Democracy Index regime type (full democracy = green, flawed democracy = blue, hybrid regime = orange, authoritarian = red); squares mark countries analyzed in this paper but absent from prior network studies. Three components emerge with 311 edges: a main cluster of 34 countries, Burkina*

*Faso as an isolate (tone = −4.95, Goldstein = −1.04), and a near-isolate Venezuela (tone = −3.90, Goldstein = −0.09). Regime type sorts vertically along the tone axis. Persistent homology confirms the topological distinctiveness of the pre-2012 and post-2016 regimes: the pre-2012 regime exhibits more persistent $H_1$ features (max lifetime 0.262 vs. 0.126 post-2016), indicating more structured coverage patterns (see Supplementary Figure S4).*

Figure 4 places countries in their tone×Goldstein position and draws edges between countries within standardized distance ε = 1.0. Three connected components emerge: a main cluster of 34 democracies and hybrid regimes, Burkina Faso as a complete isolate, and Venezuela in near-isolation at the bottom of the tone axis. The structural finding: regime type sorts vertically along the tone dimension. Full democracies (green) concentrate in the upper band (tone near −1 to −1.5); authoritarian regimes (red) concentrate in the lower band (tone near −3 to −5). Critically, this vertical sorting is not captured by the horizontal Goldstein dimension—the events themselves aren't necessarily more conflictual in authoritarian regimes; rather, they are framed more negatively regardless of content.

The graph reveals surprising neighborhoods. Israel and Iraq are connected—they sound alike in GDELT despite being a flawed democracy and authoritarian regime respectively. This is not incoherent: both have coverage dominated by conflict events in volatile regional contexts. Their similarity in Layer 2 obscures their fundamental political difference—a difference that the integration with the Democracy Index (next subsection) makes explicit. The United States, Colombia, and United Kingdom occupy the densely-connected center of the democratic cluster, near the Netherlands and Jamaica. Venezuela, by contrast, sits near-isolated at the bottom—consistent with its status as a fully authoritarian regime that has produced a uniquely negative coverage profile, structurally different from every other case in our 40-country sample.

### *Layer 3: The Dynamics of Media Deterioration*

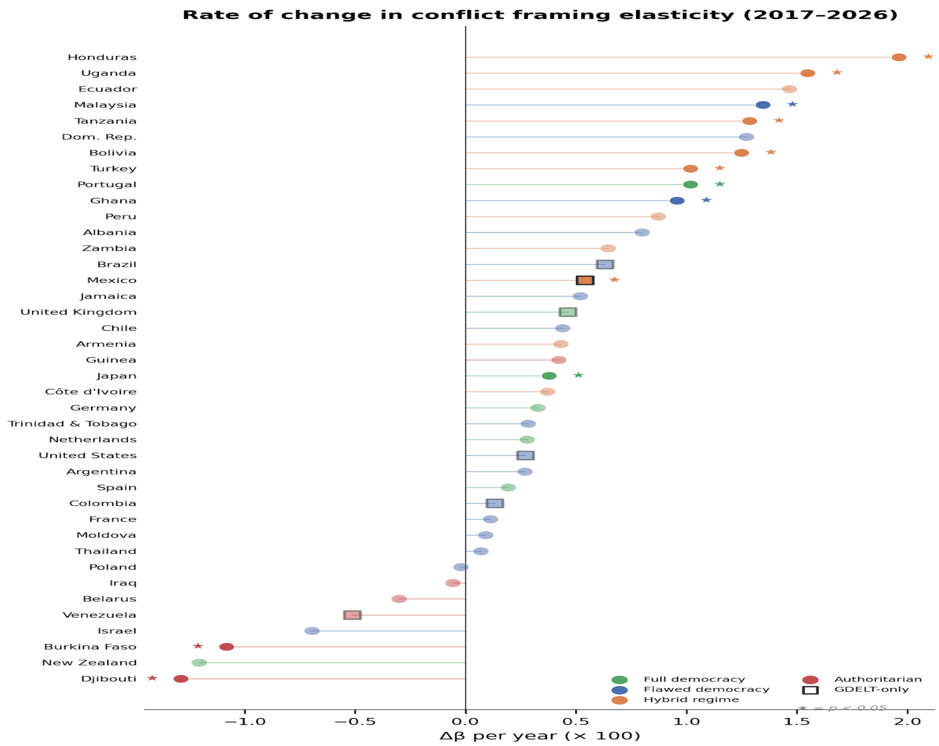


***Figure 5.*** *Rate of change in conflict framing elasticity (Δβ per year, ×100) for 40 countries, 2017–2026. Stars indicate p < 0.05. The fastest-rising β values are concentrated in hybrid regimes (orange) and flawed democracies (blue), not in the most democratic or most authoritarian regimes. Squares mark the six countries (Venezuela, Colombia, US, Brazil, Mexico, UK) that complete the regional and democratic-quality coverage of the sample.*

The most politically consequential finding of the paper appears in Figure 5. When we compute the rate of change in β across 2017–2026 for each country, the countries deteriorating fastest are not the most authoritarian; they are hybrid regimes and flawed democracies (Table 2). Honduras (+0.0196/yr, p = .016), Uganda (+0.0155/yr, p = .001), Ecuador (+0.0147/yr, p = .067), and Malaysia (+0.0135/yr, p = .008) lead the list. These are exactly the countries Huntington would identify as the contested zones of the third wave—not its strongholds or its absolute reversals, but its uncertain margins.

***Table 2***

*Fastest-Deteriorating Media Environments (Δβ/yr, 2017–2026)*

| Country | Regime Type | Δβ/yr | p-value |
|---|---|---|---|
| Honduras | Hybrid regime | +0.0196 | .016 |
| Uganda | Hybrid regime | +0.0155 | .001 |
| Ecuador | Hybrid regime | +0.0147 | .067 |
| Malaysia | Flawed democracy | +0.0135 | .008 |
| Tanzania | Hybrid regime | +0.0128 | .047 |
| Bolivia | Hybrid regime | +0.0125 | .011 |
| Turkey | Hybrid regime | +0.0102 | .006 |
| Mexico | Hybrid regime | +0.0054 | .026 |

Authoritarian regimes exhibit either stable or declining β: Djibouti (−0.0129/yr, $p = .017$), Burkina Faso (−0.0108/yr, $p = .024$), Iraq (−0.0006, ns), Belarus (−0.0030, ns), Venezuela (−0.0051, ns). This pattern is consistent with two interpretations that are not mutually exclusive: floor effects (countries whose media environments are already maximally negative cannot deteriorate further), and stabilization under authoritarian consolidation (state control over media produces uniform framing that does not respond to changes in event intensity). In either case, the dynamic signal is strongest in the middle of the democratic spectrum.

***Table 3***

*Three-Layer Profile for Major Western Hemisphere and English-Speaking Cases*

| Country | Tone | β (mean) | Δβ/yr | Note |
|---|---|---|---|---|
| United Kingdom | −1.26 | 0.264 | +0.0046 (p=.07) | Borderline rising |
| United States | −2.09 | 0.292 | +0.0027 (ns) | Stable despite turmoil |
| Brazil | −1.60 | 0.264 | +0.0063 (p=.08) | Borderline rising |
| Mexico | −2.31 | 0.259 | +0.0054 (p=.03) | Significant rise |
| Colombia | −2.11 | 0.242 | +0.0013 (ns) | Stable |

| | | | | |
|---|---|---|---|---|
| Venezuela | −3.90 | 0.191 | −0.0051 (ns) | Authoritarian floor |

Table 3 highlights the six familiar cases that may interest readers most. Mexico is the only Western Hemisphere case with statistically significant rising Δβ (+0.0054/yr, $p = .026$), placing it alongside Honduras, Ecuador, and Bolivia in the regional canary group. Brazil and the United Kingdom show borderline rising trends ($p \approx .07$–$.08$). The United States result is striking: despite substantial public perception of intensifying media polarization, β itself has not changed significantly over 2017–2026 (Δβ = +0.0027, $p = .18$). One interpretation is that the US shift may reflect changes in which outlets dominate coverage (source composition) rather than how conflict is framed when it occurs—a hypothesis that warrants future investigation. Colombia is essentially stable (+0.0013, ns). Venezuela's slight decline is consistent with the authoritarian floor pattern.

***Independence from the Democracy Index***

To test whether the three layers carry independent information, we correlate each with the EIU Democracy Index 2024 across 40 countries. Static β correlates weakly with democratic quality (Spearman $\rho = -0.13$, $p = .43$). The rate of change Δβ is independent of democracy ($\rho = -0.06$, $p = .72$). Layer 2 position (mean tone) shows the strongest relationship—reflected in the vertical sorting visible in Figure 4—consistent with the observation that authoritarian contexts produce more uniformly negative coverage.

This pattern of partial independence is substantively important. The Democracy Index measures formal political institutions and civil liberties, updated annually with substantial lag. Δβ measures the rate at which media framing is becoming more conflict-reactive, and can be computed in near-real-time from GDELT. The two metrics evidently capture different things: a country can have a stable Democracy Index score while its media environment is rapidly deteriorating (Honduras, Uganda), or a low Democracy Index score with a stable or improving media environment (Iraq, Belarus). Combining the two provides a more complete diagnostic than either alone.

## Discussion

### *The Artifact as Contribution*

Documenting the GDELT discontinuity is a substantive contribution beyond the methodological. GDELT is among the most cited open datasets in computational social science, yet its cross-version comparability has not been formally assessed. The UK Office for National Statistics flagged it as “experimental” (ONS, 2020); Ward et al. (2013) identified discrepancies with ICEWS; Hanna (2014) found inconsistencies with the Dynamics of Collective Action dataset. Several recent studies silently avoid the discontinuity: Gao et al. (2022) begin their financial sentiment sample from mid-2016. Our synchronicity diagnostic provides the first formal characterization—a reproducible test any researcher can apply. The recommendation is clear: studies using GDELT tone should restrict to a single version or apply within-year normalization, and researchers should be aware that aggregate tone trends spanning the v1/v2 boundary cannot be interpreted as substantive media change without correction.

### *β as a Diagnostic*

The conflict framing elasticity captures something that raw tone does not: the reactivity of a media environment to the conflict dimension of events. The Venezuela–United States inversion (most negative coverage / lowest β vs. near-average coverage / highest β) distinguishes ambient negativity from conflict reactivity. Venezuela’s uniformly negative coverage is the condition Arendt identified as the precursor to epistemic collapse: when everything is crisis, nothing is distinguishable. The United States' high β indicates a different pathology: a media environment hyper-reactive to conflict that amplifies the tonal contrast between cooperative and conflictual events—consistent with the perception that political conflict is more intense than the Goldstein scale shows it actually is.

### ***The Canary Cases***

The dynamic analysis (Table 2) identifies the countries where the third wave's reversal is being measurably tracked in real time: Honduras, Uganda, Ecuador, Malaysia, Tanzania, Bolivia, Turkey, and Mexico. These are hybrid regimes and flawed democracies—precisely the contested middle of the democratic spectrum. Their media environments are becoming more conflict-reactive at statistically significant rates, even as the Democracy Index shows more gradual changes. β appears to be a leading indicator of democratic backsliding, operating on a faster time scale than formal regime classifications.

This has important implications for the Arendtian argument. Arendt's warning was about the structural conditions that enable authoritarian capture—the dissolution of shared factual reality. The countries deteriorating fastest in our data are not the ones that have already failed the Arendt test; they are the ones whose media environments are beginning to drift toward the conditions she warned about. This is consistent with Diamond's (2015) observation that the democratic recession is defined less by outright collapses than by gradual, cumulative erosion.

### ***Layer 2 and Regime Sorting***

The vertical sorting of regimes in tone×Goldstein space (Figure 4) is a distinct finding. Authoritarian regimes cluster at the bottom of the tone axis—their coverage is uniformly more negative across all types of events. This is not a property of β (which varies independently of regime) but of mean tone. The mechanism is likely source composition: in authoritarian or crisis contexts, the news pipeline is dominated by conflict stories, negative international coverage, or state-controlled media framing that does not distinguish between cooperative and conflictual events. Venezuela's near-isolation at the bottom of the skeleton is the strongest expression of this pattern—a country whose media narrative has become structurally different from every other case in our sample.

### *Implications and Future Work*

The three-layer framework suggests several practical applications. β can be computed in near-real-time from GDELT's 15-minute update feed, providing election commissions, civil society organizations, and democracy monitoring institutions with a media-side indicator that complements existing political indices. The synchronicity diagnostic provides a standard tool for assessing measurement artifacts in similar datasets. The Δβ measurement, when applied to the canary countries identified here, could be combined with downstream measures of conversational structure (e.g., social network polarization indices) and political outcomes (electoral integrity, protest activity) to test whether Δβ indeed leads observable democratic deterioration. The relationship between media β and the network-level measures of discourse fragmentation analyzed in our prior work (Cárdenas-Sánchez, Sampayo, Rodríguez-Prieto, & Feged-Rivadeneira, 2022) is a natural follow-up question for a separate study.

### *Limitations*

GDELT tone derives from automated sentiment analysis with potential cross-language biases. The z-score correction assumes relative positioning is comparable across regimes. The β level-shift restricts trend analysis to a decade. The source-composition hypothesis for the discontinuity has not been tested through source-level filtering. The Δβ analysis assumes linearity; some trajectories (Ecuador, Honduras) show non-monotonic patterns. The 40-country sample, while broader than typical comparative studies, is not exhaustive—notable absences include China and many small democracies. The framework as presented measures media coverage, not media production or audience reception; complementary measures of those dimensions would strengthen causal inference.

## Conclusion

We have proposed a three-layer diagnostic framework for media coverage in democracies: conflict framing elasticity (β), position in the tone×Goldstein coverage landscape, and rate of change (Δβ). Across 40 countries, the framework identifies the canary cases of democratic erosion: hybrid regimes and flawed democracies—Honduras, Uganda, Ecuador, Malaysia, Tanzania, Bolivia, Turkey, Mexico—where media environments are deteriorating at statistically significant rates. Authoritarian regimes are stable at the floor of the measure; full democracies show no significant deterioration over the observation window. The deterioration is concentrated in the middle of the democratic spectrum, where political contestation is most intense and outcomes are least determined.

The GDELT discontinuity, while an artifact, provided the methodological foundation for this framework. By forcing us to treat pre-2012 and post-2016 as separate measurement regimes, it produced a decade of consistent data within which genuine trends can be assessed. The synchronicity diagnostic offers a reusable tool for characterizing similar artifacts in other large-scale media datasets.

Three successive media revolutions—radio, television, algorithmic platforms—have each disrupted democratic discourse. Each of the first two was eventually brought under institutional governance. The third remains largely unregulated, and our data suggest its consequences are still unfolding. The countries whose media environments are deteriorating fastest are precisely those that Huntington's third wave brought into the democratic fold in the 1980s and 1990s. Whether they complete the reversal Huntington feared will depend in part on whether the structural conditions for democratic communication can be restored.

Arendt warned that democracy depends on a shared world. The data in this paper suggest that world is under measurable strain along multiple independent axes, and that the strain is accelerating in the places most politically vulnerable to it. The tools we offer can detect the strain in near-real-time. Repairing it requires political will operating on longer timescales than media.

## Declarations

### *Data Availability*

All GDELT data analyzed in this study are publicly available via Google BigQuery (project: gdelt-bq, table: full.events). The EIU Democracy Index 2024 scores are published by the Economist Intelligence Unit. BigQuery SQL queries, Python analysis scripts, and figure-generation code are available at https://github.com/alejandrofeged/conflict-framing.

### *Code Availability*

All code used to produce the analyses and figures in this paper is available at https://github.com/alejandrofeged/conflict-framing under an MIT license.

### *Competing Interests*

The author declares no competing interests.

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### *Author Contributions*

A.F.-R. conceived the study, designed the methodology, collected and analyzed the data, produced all figures and tables, and wrote the manuscript.

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